

Practitioner's Commentary: The Outstanding Drug Runner Papers

Walter Stromquist

Daniel H. Wagner Associates
Station Square Two
Paoli, PA 19301

I liked the comment of the team from the North Carolina School of Science and Mathematics about not expecting the searchers to have an advanced computer when they can't even afford radar. An actual search under these conditions would almost surely use radar, so this problem is somewhat artificial. Also, the distances mentioned in the problem are somewhat unrealistic: A helicopter can be heard for miles, and a good searchlight can illuminate a boat in open ocean at a distance of a mile or more—at least well enough to make the searchers come back for a closer look.

I suspect that this problem has been contrived as a proxy for a much more realistic problem in antisubmarine warfare. For example, the target might be a lurking submarine, and the searcher might be a destroyer using active sonar. The destroyer sends out “pings,” which will allow it to detect the target at a certain distance; but the pings can be heard by the submarine at a greater distance. The submarine is effectively much slower than the destroyer, because if the submarine attempts to go too fast, the destroyer's sonar operators will hear the sound of the submarine's engines. Alternatively, the searcher could be another submarine. Searches of this kind arise repeatedly, so even if the problem is artificial in the form presented, it is not irrelevant.

Problems like this are called (not surprisingly) *searcher path problems*. From a theoretical point of view, they are notoriously difficult—in fact, some versions of the problem are NP-complete. Therefore, we are not likely to find any efficient mathematical algorithm to compute a provably optimal path. In practical situations, there is no better approach than the one chosen by both of the Outstanding teams: Think of some candidate search plans, evaluate each plan on the basis of some model, and pick the best of the candidates. There is never any assurance that we have thought of the best plan.

Still, there are some analytic techniques available for dealing with these problems. I will describe some of them under these headings:

- the probability distribution;

- sweep width; and
- candidate search plans.

Then I'll comment on wake detection and other topics.

The Probability Distribution

Both teams did a fine job of describing the target's initial location. Both teams assumed (as I would have) that each of the bearings is subject to a normally distributed error with standard deviation 1° . They could therefore draw a wedge around each bearing line, extending 2° to either side of the line, such that each wedge has a 95% chance of containing the target. The intersection of the wedges is a "search area" roughly 35 mi long and 2.5 mi wide at its widest point, with area about 40 sq mi. Note that the probability that the target is actually in this area (i.e., in both wedges simultaneously) is about 90%.

Both teams recognized that the search area is really only a convenient way of representing a complicated bivariate probability distribution, and both teams found effective graphical ways of illustrating the distribution.

What if the target has been moving since the message was intercepted? In that case, the search is probably hopeless, so we may as well assume (as both teams implicitly did) that the target is stationary.

Sweep Width

We can model the searcher as a broom, sweeping a path through the search area and finding anything within the path. The shape of the broom, or *catch zone*, is unimportant; all that matters is its width, which we call the *sweep width*. It is one of the most important concepts in search theory, because it summarizes the capabilities of the sensor in a single number. If we aren't sure of the number, we can vary it parametrically and, in effect, test the sensitivity of all of our sensor-related assumptions at once. Both teams came up with numbers for the sweep width. In the simplest cases, the North Carolina team used 400 ft (the diameter of a circle), and the Drake University team used 283 ft (the width of the forward-looking 90° wedge).

Neither team really came to grips with the use of the searchlight. If the helicopter is traveling at 140 or 180 mph, then it will be impossible to swing the beam back and forth fast enough to find all targets in a 283-ft or 400-ft path. Even if it takes only 2 sec to sweep from left to right and back to the left again, the helicopter will have traveled at least 400 ft, leaving the left side of the path uncovered for most of this time. For part of the cycle, the searchlight will be centered, making the sweep width nearly zero. Perhaps

the best strategy is to keep the beam pointing to one side, at right angles to the path; that would guarantee us a sweep width of 200 ft.

Both teams devoted substantial effort to modifying the sweep width based on the target's evasion strategy. This led to some elegant geometry, but it was probably a mistake, given the distances and speeds in the problem. Once the helicopter has come close enough to be heard by the boat, it will cover the 500 ft between itself and the boat in less than 3 seconds, giving the boat no time to react. It is better to assume that the boat is stationary, and that any evasion strategy takes effect only after the first encounter is complete. This leaves the sweep width unaffected by the evasion strategy, considerably simplifying the problem.

If we ignore the "scare zone," then we can use the sweep width to obtain a simple estimate of the ultimate success probability of the search. Multiply sweep width by searcher speed to get *sweep rate*, in units of area covered per unit time; then multiply by the time available to get the total area that can be covered. Using the North Carolina team's numbers, the calculation is even easier: The length of the search path is assumed to be 225 mi., so that the area that can be covered is $(400 \text{ ft}) \times (225 \text{ mi}) = 17.0 \text{ sq mi}$. This is less than half of the search area; but by covering the 17 sq mi of highest probability, we can obtain a success probability of 50–60%. This is almost without regard to the path chosen: All that is necessary is that we stay in the high-probability area and avoid overlapping previously searched territory.

But we cannot ignore the scare zone. A better model is that the path of the broom is 1,000 ft wide, as in **Figure 1**.

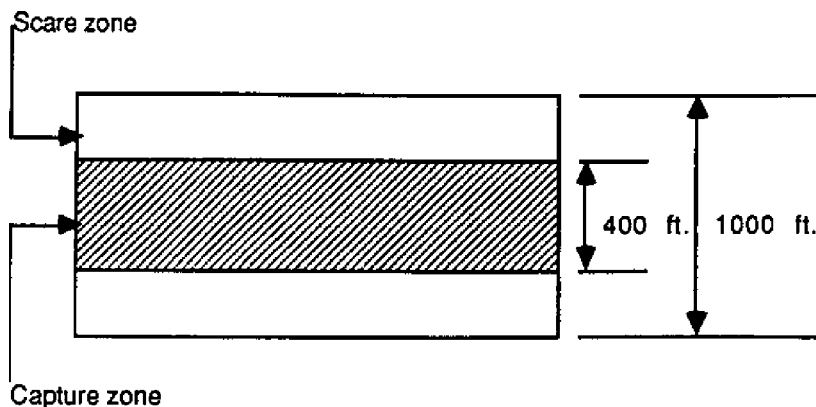


Figure 1. Model of the searchers' path.

Outside the capture zone (say 400 ft wide) is a space in which the target will hear the helicopter and presumably escape before the helicopter returns. With a sweep width of 1,000 ft, we can cover 43 sq mi in the time allowed; but even if the target is in the area covered, we have only a $400/1000 = 40\%$

chance of capturing it. It is better to use the North Carolina team's idea of spacing the tracks as in **Figure 2**, so that one side of the scare zone overlaps previously covered territory. Now our effective sweep width (sweep width in new territory) is 700 ft, so we can cover about 30 sq mi, and the chance of capturing the target given that it is in the covered area is $400/700 = 57\%$. More accuracy than this requires a simulation.

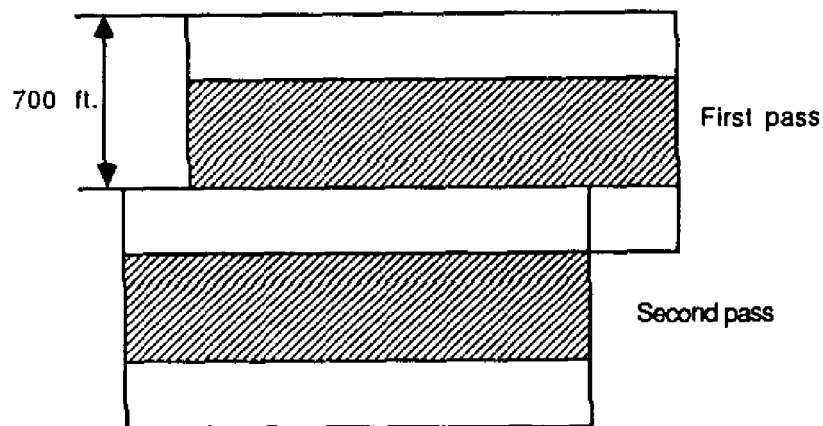


Figure 2. Overlapping paths.

Candidate Paths

The argument from sweep width shows that the exact path chosen makes little difference, as long as the path stays within the highest-probability area and the track spacing is such as to avoid unwanted overlap.

If the target is not expected to escape (or if it is expected to return immediately, which amounts to the same thing), then the appropriate track spacing is equal to the sweep width. This avoids overlap of coverage while still packing the tracks as tightly as possible into the high-probability area. The Drake University team's simulation results bear this out: In the cases with an escape-and-return strategy, the smaller track spacings were most effective. The results also show that track spacings of 500–1000 ft are most effective if the target is prone to escape permanently; this is consistent with the plan of **Figure 2**.

Aside from these considerations, any preference for one path over another has to be based on something more subtle than our simple sweep-rate model. For example, both teams were quite ingenious in designing paths that maximized the probability of catching the target in the course of its escape. This is good thinking, but there is no way short of a simulation to test the effectiveness of their ideas.

Another consideration involves timing. There is the constant risk that the target will complete its business and leave the area. Therefore, it is better, all other things being equal, to search in the highest-probability areas first. This is an argument for the “outward” patterns (in the Drake University team’s scheme) that isn’t addressed in their simulation.

Wake Detection and Other Topics

Some readers may be surprised to learn that the North Carolina team’s assumptions about the persistence and detectability of the boat’s wake are very reasonable. A fast-moving power boat in open ocean may indeed leave a wake that can be detected from the air, even by searchlight, for as long as half an hour. The wake makes itself more visible by stirring up luminescent plankton. Of course, this depends on the weather and on the size and speed of the boat.

The North Carolina team also proposed a very clever method of taking advantage of the wake: by using the same track spacing as in **Figure 2**, they ensure that if the boat is “scared” on one pass, it will almost surely leave a wake across the path of the helicopter’s next pass. After spotting the wake, the helicopter can quickly run down the target. In effect, the team has gotten the advantage of a 700-ft sweep width, without leaving gaps in coverage.

I would recommend two changes in the North Carolina team’s final plan. First, their back-and-forth plan calls for the helicopter to make a U-turn every 30 to 60 seconds, which is probably impractical. It would be better to lay the tracks out along the length of the search area, even though the wake would have more time to age between one pass and the next. Also, as noted above, I would try to search the highest-probability area first, even at the expense of an extra transit or two. My recommendation is therefore something on the order of **Figure 3**. In fact, that would be my recommendation whether wakes are detectable or not. I do not know whether this plan is really better than the North Carolina team’s plan, or the best of the Drake University team’s quadrangular plans.

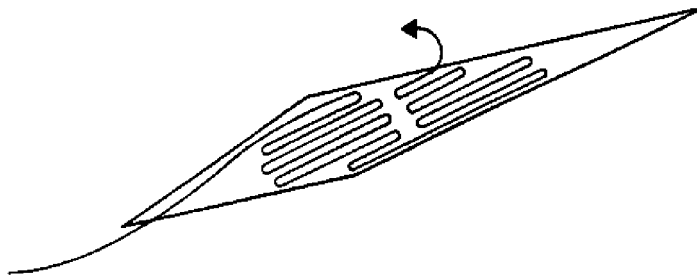


Figure 3. Recommended plan (not to scale).

Some idle questions: If the inference in the problem is correct, won't there be *two* boats in the area? If wakes are detectable, wouldn't the target be better advised to remain stationary?

The North Carolina team poses a question for the helicopter that might be asked of a dog chasing a truck: When you catch it, what are you going to do with it? In fact, there are ways for a well-armed helicopter to apprehend a boat: either lower a boarding party, or escort the boat to port under threat of sinking. Both methods take time, which must be allowed for in planning the search.

The Drake University team is to be congratulated for their simulation program. It is exactly the tool needed for this kind of problem, since it can be adapted to almost any set of assumptions about searcher and target strategies. To bring such a program to the point of producing usable results in the limited time available is a remarkable accomplishment.

An excellent reference on search theory is Stone [1977].

Acknowledgment

These comments have been prepared with the assistance of Dr. Joseph P. Discenza.

Reference

Stone, Laurence D. 1977. Search theory: A mathematical theory for finding lost objects. *Mathematics Magazine*. 50 (5) (November 1977): 248–256.